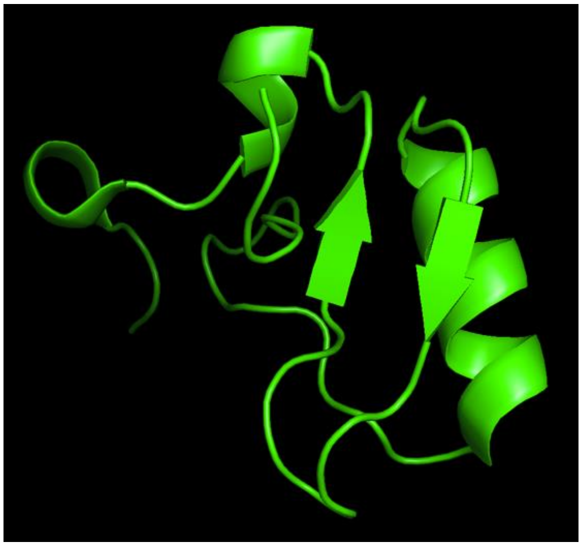
Amino Acid Properties in Observed Structural Motifs

Structural Analysis of Scorpion Toxin Protein (PDB: 2I61)

Inna Kucherova

Part I: Structure Quality & Metadata



1. PDB Information

- **PDB ID:** 2I61
- **Title:** Crystal structure of the scorpion toxin LqhIT2
- **Organism:** *Leiurus quinquestriatus* (Deathstalker scorpion)
- **Year:** 2006
- **Study:** https://doi.org/10.1016/j.jmb.2006.10.085

2. Methodological Quality

- **PDB ID:** 2I61
- **Resolution:** 1.20 Å (Atomic Resolution)
- **R-values:**
  - **R-Value Work**: 0.096 (9.6%)
  - **R-Value Free**: 0.120 (12.0%)

Structure of scorpion toxin was received via X-ray diffraction. High resolution of 1.20 Å allows clear visualization of atoms and side-chain orientation of molecule. Free R-value 0.120 is significantly low (usually <0.20 is very good), so we can conclude high stability of the toxin model and absence of overfitting.

3. B-Factor Analysis

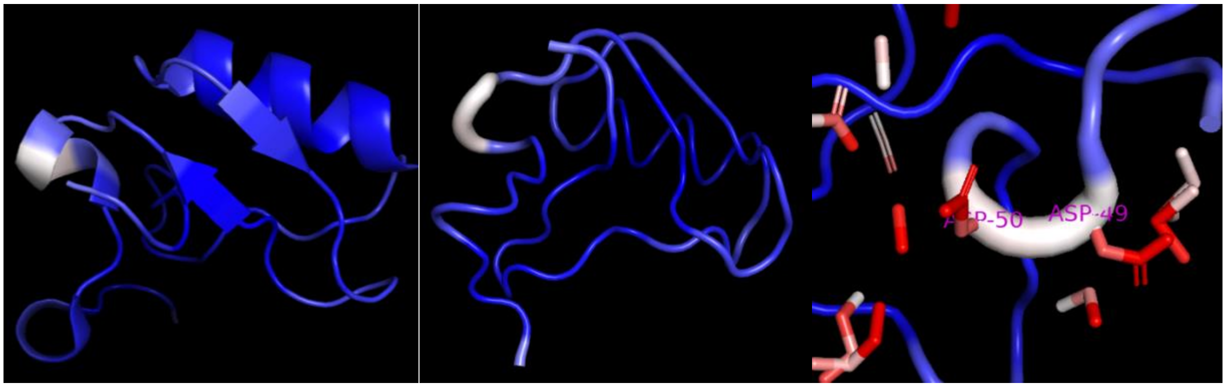
Like many other toxins, PDB 2I61 has a highly rigid structure, which makes it stable in the environment and effective in its toxicity function. For best visualization of toxin flexibility the putty mode with threshold set at 25 was used due to precise and rigid structure of molecule.

- **The Rigid Core:** alpha-helix and beta-sheets are colored in deep blue which means low B-factor with minimal fluctuations of atoms, which can be explained by disulfide bridges connecting the protein structure together.

- **The Flexible Loop:** scorpion toxin has only one flexible loop in C-terminal part, containing Asp-49 and Asp-50. Two negative charged residues of aspartic acid in this loop allow toxin functions like:

- **Maximum solubility** in insects (or other species) blood, by forming dense hydration shell with water molecules around loop and protect from protein aggregation
- **Flexibility** by repelling of charged side chains of Asp-49 and Asp-50 attend to water, so this mobility allow to adapts to the receptor's shape
- **Electrostatic steering** for attacking sodium channels in the nervous system by creating a local electric potential via negative charge of the Asp-Asp pair and significatly accelerates binding poison to receptor

According to Karbat et al. this flexible loop of the C-terminal segment is critical for receptor interaction, allowin toxin adapt to sodium channel binding pocket.



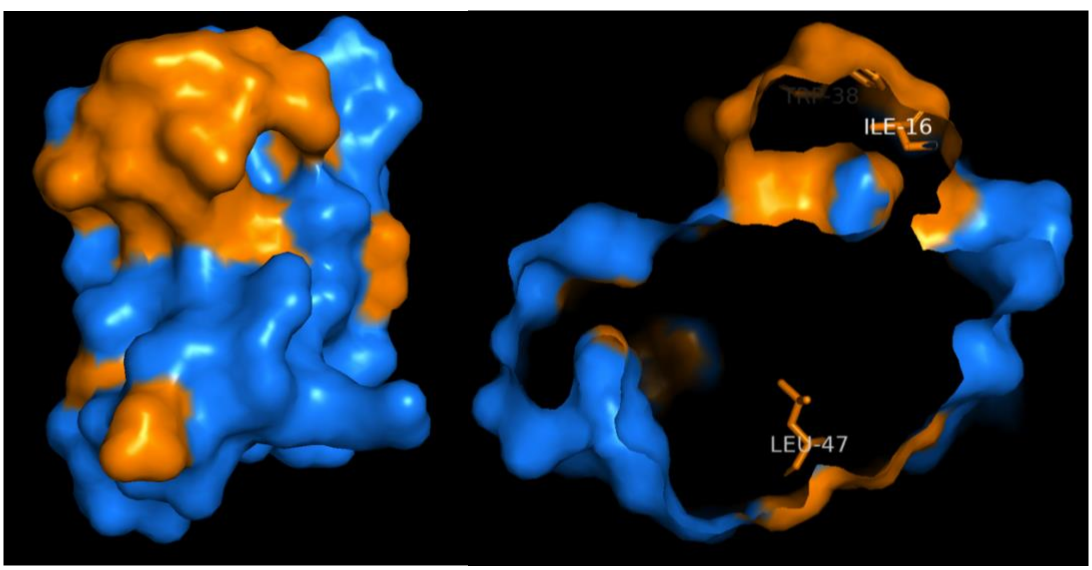
Part II: The Hierarchy of Structure

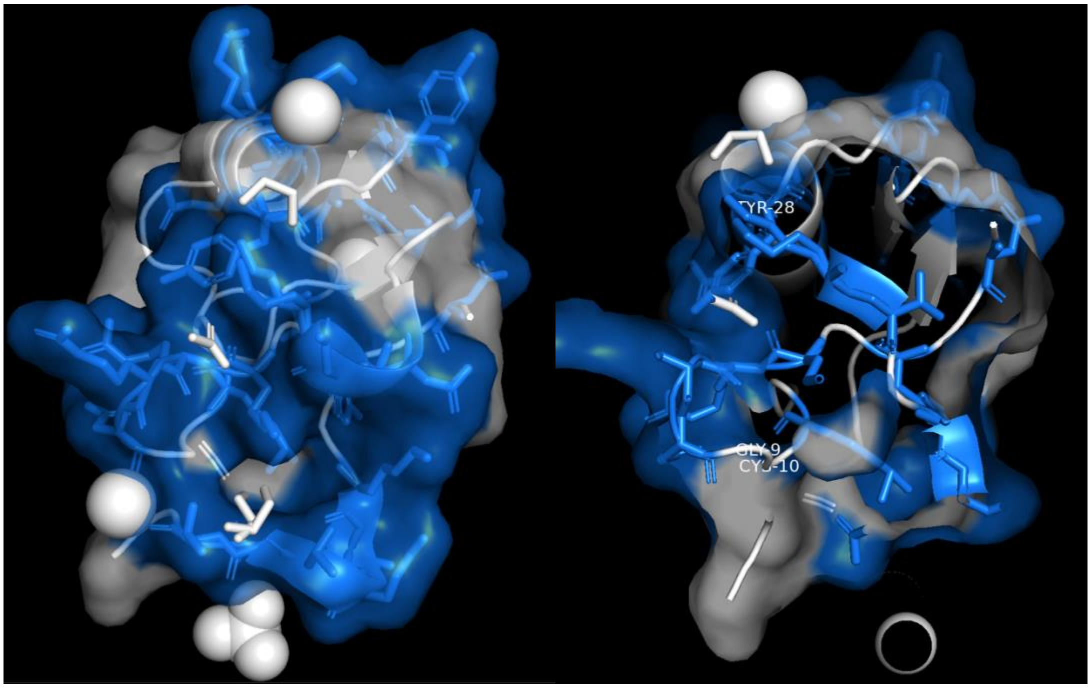
2. Hydrophobic Core vs. Solvent-Accessible Surface

2i61 molecule has a classic hydrophobic core formed by residues Trp-38, Leu-47, and Ile-16. Trp-38 is central amino acid of core stability, his indol ring interacts with Leu-47 and Ile-16. Mutations in this position for other similar toxins leads to complete loss of structure, so we can conclude that Trp-38 here plays key role as structural anchor.

On the other side molecule has solvation shell where water molecules requruited by flexible loop interact. Labeled residues like Tyr-28, Gly-9 and Cys-10 are located on the outer side of the molecule, reaching out into solvent space. Tyr-28, even though it has a hydrophobic ring, polar -OH making it suitable for surface. Gly-9-Cys-10 loop performs structural function:

- Cys-10 forms a conservative disulphide bond, which stitches N-terminal loop to the hydrophobic core and limits its mobility
- Gly-9 acts like pivot due to its conformationand allows the polypeptide chain to adopt the needed curvature to bind cysteins together





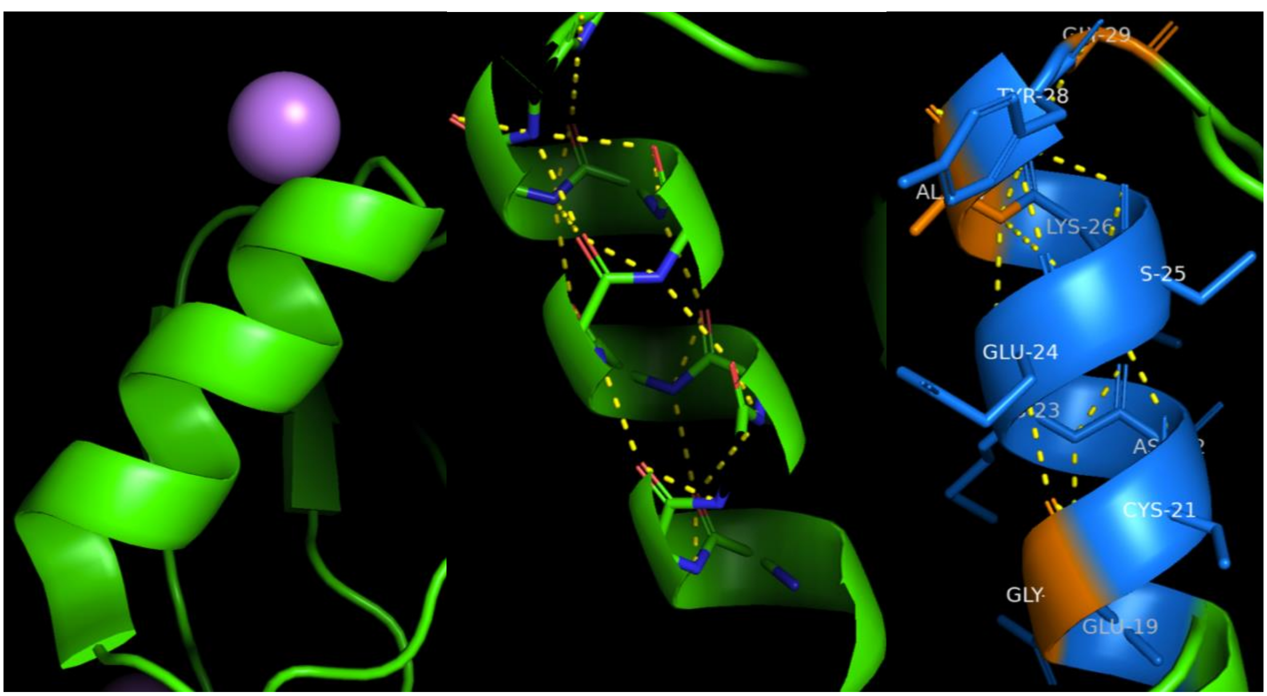
Part III: Analysis of Secondary Structural Motifs

2i61 molecule has classical for scorpione toxins folding – cysteine-stabilized alpha-helix and beta-sheet. Only one alpha-helix (residues 19-29) is packed parallel to a double antiparallel beta-sheet. Structure held together with a hydrophobic core and disulfide bridges, which connect the alpha-helix directly to the beta-sheets.

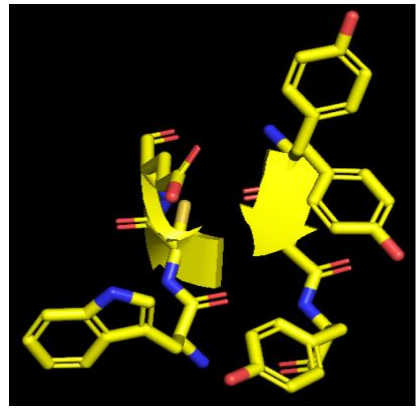
1. The Amphipathic -Helix (Residues 19–29)

Alpha-helix demonstrate clear periodicity with 3.6 residues per turn, which creates functional edges:

- Hydrophobic edge – like Gly-20, Ala-27 – oriented inside protein forming contact with the beta-sheet
- Hidrophilic edge – like Glu-19, Glu-24, Lys-26, Tyr-28 – alighn to the opposite face forming hydrophilic side



2. The Anti-Parallel -Sheet

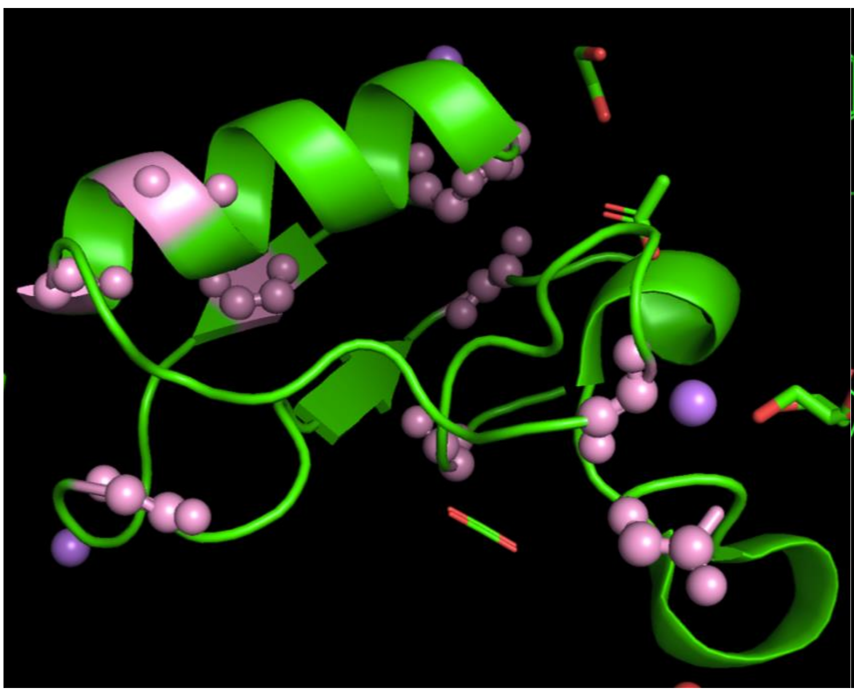


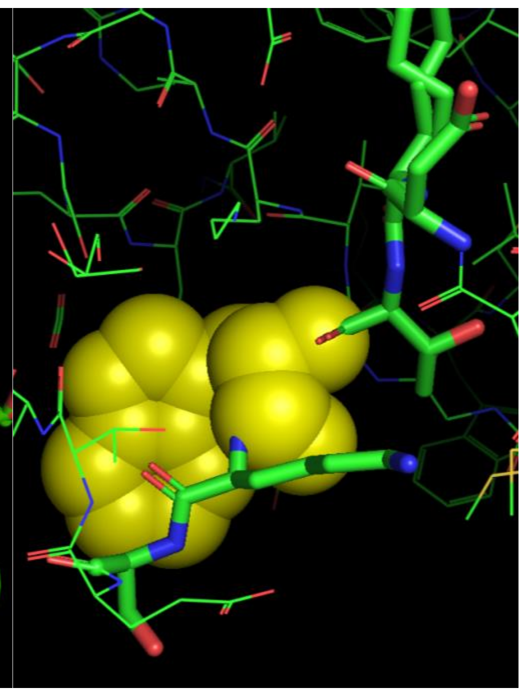
Two antiparallel-oriented beta-sheets are evidenced by the opposite direction of arrows. The side chains have classic zipper packing where hydrophobic residues (like aromatic rings) from oppositing strands alternate and connect to each other. So, this tight packing allow to stabilizes beta-sheet and forms a hydrophobic core.

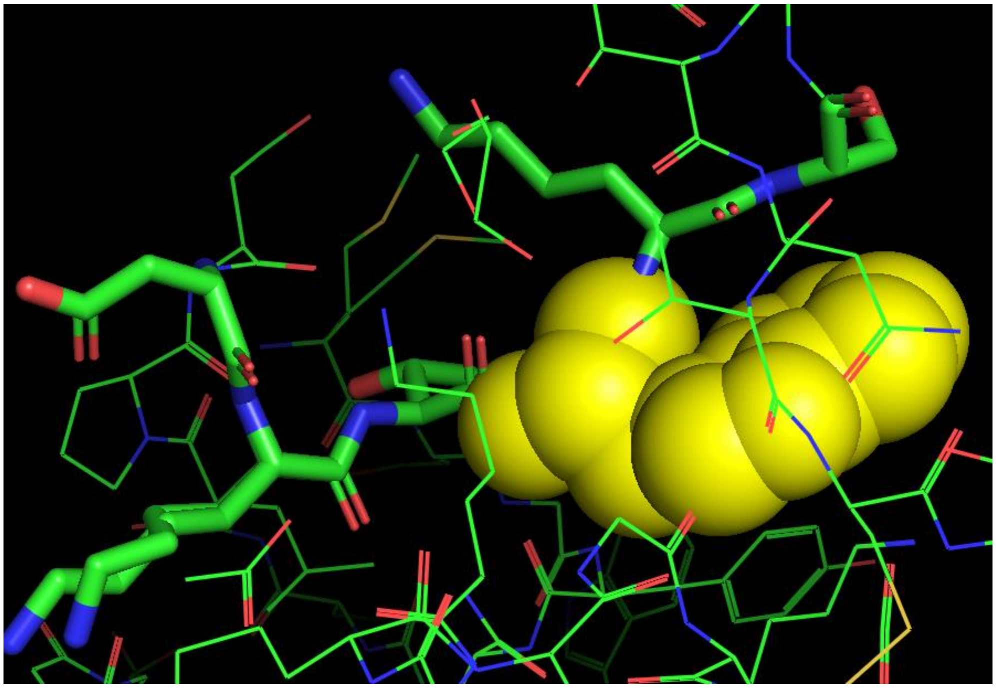
Part IV: The Structural "Rebels" (Glycine & Proline)

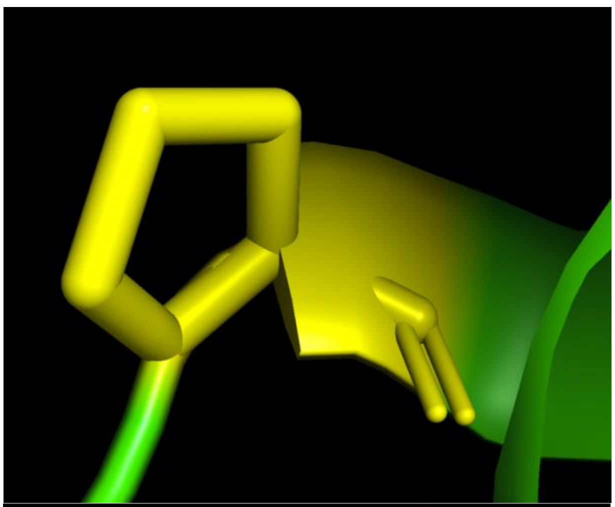
3. Glycine-53

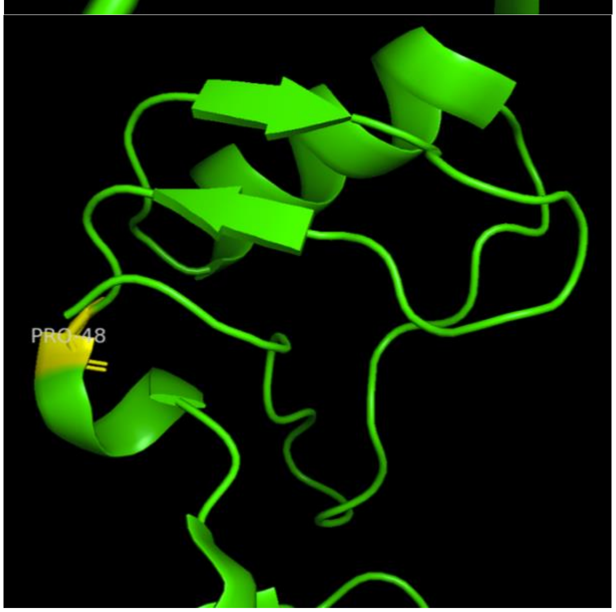
Gly-53 has a torsion angle phi -176.70 , so the tail or protein no longer twisted into loop and becomes extended. C-terminus of scorpion toxin (50-61 residues) – long segment with some flexibility, not glued to core, and sticking out like lash. For neurotoxins, like 2i61, this flexibility can be advantage – like flegible loop described above, it can be used as tool for searching target. And once toxin find sodium channel – this C-terminus acts like anchor – enters the receptor pore and becomes fixed there.











4. Proline-48: The "Helix Breaker"

Pro-48 was identified by unique cyclic side chain – pyrrolidine ring. Proline acts like helix breaker by two factors:

- Loss of amide hydrogen on the N-atom pprohibits the formation of hydrogen bond
- Rigid ring limits conformational freedom, fixing torsion angle $\sim -60^{o}$

So, geometric kink forms in the polypeptide chain which leads to secondary structure trunktation or rigid turn formation.
Pro-48 located right in front of flexible Asn49-50 residues, its likely pushes the loop into the surface at a certain angle, ensures that loop will be extend outwards. Bonds created by proline are hard to break down by regular enzymes (eg proteases), so placing Pro next to the active loop – scorpion protects toxin from rapid destruction.

Conclusion

2i61 is a good evolutionary design example where each deviation from standard geometry and tiny size is not a bug but feature for high afinnity and toxicity of protein.

reference: https://doi.org/10.1016/j.jmb.2006.10.085